SIMULATED / PUBLIC EVIDENCE

A sealed cohort. One public result. Round 2

Real cryptographic checks over synthetic inputs, actors, keys and a declared sector denominator. No accreditation, official acceptance or measured climate impact is asserted.

0.848636

kg CO2 per kg product - programme boundaries must be reconciled before official use. The displayed ratio is rounded down to six decimals. Badges compare exact rational thresholds.

Public field	Value
Aggregate emissions / kg CO2	2079159525
Aggregate production / kg	2450000000
Contributing installations	5
Declared sector production / kg	9187500000
Declared-denominator coverage / ppm	266666
Protocol representative flag	False

Pinned public evidence

Chain 10143 / block 73911871 / block time 2026-10-24T20:48:01+00:00

`Round contract: 0x9ea9702324c53dcfb3c179acfed122dbf52f9cfc`

`Block hash: 0x01f659fa991ce704a79103e2061804ea28838eded3c547ee4b0fa79c9c2fceb7`

`Frozen manifest: 0xe65b4fbe3cc46f847866e06ca67bbf56707b522249bbaada48194a967894bbff`

`Canonical payload SHA-256: 0xb3a976a01f890557034e5b5aa42c0f99ff25754978feee2f2fca37aa54872efe`

This PDF is a readable summary. The companion JSON contains the complete commitments, trustee publications, signature sidecars, parameters and events. The checksum detects changes to that payload; it is not a regulator signature.

Reproduce the checks

1. Obtain the source build and expected deployment anchor independently. A file supplied beside this dossier is a convenience copy, not an independent trust root.

2. Validate schemas/dossier.schema.json and recompute the canonical payload checksum and source hashes.

3. Fetch the selected RPC at the pinned block, check chain ID and all four deployed code hashes, and match the complete snapshot and public events.

4. Recompute the exact ordered manifest, point sums, every published trustee VSS sum, interpolation, aggregate openings, intensity and declared-denominator coverage.

5. Check every allowlisted ECDSA/ML-DSA-65 co-signature and successful transaction receipt. ML-DSA is checked off-chain.

```
node --import tsx scripts/verify-dossier.ts --dossier SIMULATED_dossier.json --deployment
SIMULATED_deployment.json --rpc http://127.0.0.1:PORT
```

Offline mode: add --offline and omit --rpc. It verifies arithmetic and signatures without authenticating chain history. Local RPC links require access to the originating machine; no public phone endpoint is implied.

Frozen contributing installations

`1. 0xfbf398724193cc53959d2e2c6f51827849e5f663ff83d3e8b9412c2afe6cadd5`

Complete public commitment and ciphertext hashes: JSON /payload/snapshot/plants/0. No individual opening is included.

`2. 0xef53f13f4a104912d6d4b8662852a1d139fb5bfc5fb551eacdee6029cf2c46ec`

Complete public commitment and ciphertext hashes: JSON /payload/snapshot/plants/1. No individual opening is included.

`3. 0xf47eaf8e280bd035035b5c7456b5f6efc41e8b8a6d2c7617c7fc133bf83e9fdb`

Complete public commitment and ciphertext hashes: JSON /payload/snapshot/plants/2. No individual opening is included.

`4. 0xd644d5b5c0487c6755b31f181c6ed5a1d963fbe33c97ea427c4e1434636cd7e0`

Complete public commitment and ciphertext hashes: JSON /payload/snapshot/plants/3. No individual opening is included.

`5. 0xf1a5e87493b586a5ccb830aeb87010257f75745c6f62309d3145d72c0bd28590`

Complete public commitment and ciphertext hashes: JSON /payload/snapshot/plants/4. No individual opening is included.

Trustee publication record

Publication mask 6 / fault mask 1. Complete public share-sum values are in JSON /payload/snapshot/sums. These are aggregate trustee shares, not per-plant plaintexts.

Successful transaction evidence

29 unique successful transaction records. A reverted attack has no successful event and is documented separately in the attack-console evidence.

Block 73906335 / RoundCreated, StateChanged `0xd1f398178410937501311fb959c64572dea76f026fe9133733b4bd007867e850`	Block 73906350 / StateChanged `0xf516841e5902fa66cc939ecd1374f48a748b2e273a9add47d3c560f7052274a4`
Block 73906444 / PlantCommitted `0xf00c70d7292f5b67b788fd4eab0d20207535cb44c2441d5e370e2d9c81895cec`	Block 73906555 / PlantCommitted `0x2db241f7fa00e60719bcff2bd2c20056e35b6dcdaa57cbbe413f38dae8c4f994`
Block 73906597 / PlantCommitted `0x5af2ce303af8e20347b28d0c53397084b8a9e17b23ccf50bbaf952bb5a03f48b`	Block 73906641 / PlantCommitted `0x360851010876ccd48c8e092ef339379b9120bf00d28d11a84f5bcf94b8aca1c7`
Block 73906684 / PlantCommitted `0xb5516a54c89fd21bce9f563e501e4a0d338c5156ba4892716c11213e497ebf28`	Block 73906712 / StateChanged `0x7ac750990c26f8d62e05c1f3e8fc1030f2e167124aa0d4e1e72adb54ce023cae`
Block 73906765 / ShareAcknowledged `0x2bda6f81632edbdad5138819d38ef5d4da360ab30e6fc2cb30c960f4871b75c9`	Block 73906780 / ShareAcknowledged `0x8014565da0e7f56570b5ee6baca0d4b1c4007c0a581a51032ca79efdd37233bb`
Block 73906801 / ShareAcknowledged `0xf14b496c75868da7faad84deafbaef4e6366b10af9aa3e1b290682844edc3453`	Block 73906853 / ShareAcknowledged `0xbb8fc9a8d0d2f6cd1fbe6aece5c68e92157e86a322a01eca0dd4e2a3095fe583`
Block 73906872 / ShareAcknowledged `0x698aa003578b7988b42c3cecac4481e718ca04d8cd8d37018fc4d68716151c33`	Block 73906897 / ShareAcknowledged `0x4f78cc765e3ac1d5649dddafb2b92a8fb1876cd92c055a1f82f985b9f1a71f70`
Block 73906944 / ShareAcknowledged `0x83681332f71bf03fcc8d4fb24a0245cb9dd338d8c54735e08b4cb2cbc9530d6f`	Block 73906962 / ShareAcknowledged `0x7f19b416692d1800caf30799084e5708ea75137ebf82959c08f487ee62875666`
Block 73906984 / ShareAcknowledged `0x87f8e7f6e305d757dea967a46c0acf29bda1a373a7e66219c4b2e171cc68fd89`	Block 73907034 / ShareAcknowledged `0x63ebf616d6d9b2ea19182abde05b3bbc7681daf8b739d7b0d5e560824eb328d9`
Block 73907065 / ShareAcknowledged `0x4a9aafd743eb977b48a804c0d30eaa0dfa2d42450d7d3b71bb57f3052c0352b0`	Block 73907085 / ShareAcknowledged `0x7d81000b01f74b98bf13ecd670ae9927fb9d77c8d164d099dbcad8c6d05f77fe`
Block 73907137 / ShareAcknowledged `0xcdeb82b293ba9148be6644dd332568ed7a76ff4ec35b63f3e2234b0ddebf409f`	Block 73907160 / ShareAcknowledged `0x93ae503e668e405bba893b0a4cf6d5617076f086bca1bb14b4a598f9e33ed5cf`
Block 73907184 / ShareAcknowledged `0x33551cd5eea1d32b072d0d08d39708bdb3a86dcf2a658e1287c08d943252394d`	Block 73907207 / ManifestFrozen, StateChanged `0x5dcb9b0212df9cf5ec1d2081ad4f954a4d1336d7c12662d61c2b03251ec44954`
Block 73907296 / TrusteeFaultRecorded `0xe0fc236ea54175d33fe9d302853e95ea4fd17f3a26ec8e7c5261cb28d265c670`	Block 73907345 / SumPublished `0xacc4c66e2dae2f38e63e51060eb92257fac879d92d8e8344bfba44baabb43c3c`
Block 73907399 / SumPublished `0xa0415ed95294483ea2bedba3933fad8d488361b8c89949077e3627d94b4c727d`	Block 73907423 / RoundClosed, StateChanged `0xe20feb622802fd81070f69f9b670fd819d4c63321c9230420f5e420758614af8`
Block 73907448 / BadgeAttested `0xd95fef4606b00db2c6a09f5aaa832747286641f7bbb908331d12361b1fc06601`	

Provenance and limits

No private report, opening, individual share plaintext, signing key, trustee storage key or job capability is included. Verifier identities are synthetic and unaccredited. Report truth and official acceptance are not established.

1. SIMULATED reports, inputs, actors, keys, trustees and sector denominator; same-host operation.
2. Arithmetic and signatures do not establish report truth, accreditation or regulatory acceptance.
3. Two colluding trustees can reconstruct individual plants; endpoint compromise can expose openings.
4. Aggregate subtraction by other plants and overlapping/repeated cohorts can disclose individual information.
5. Each issued threshold badge reveals a comparison bit; the bound is per round, not a global query budget.
6. ECDSA, Groth16, commitment binding and X25519 transport have classical assumptions; this is not a fully post-quantum system.
7. ML-DSA co-signatures are checked off-chain. The single project setup contribution has no independent production audit.
8. Coverage uses a simulated declared denominator. The representative flag is a protocol policy, not official population assurance.
9. The generic aggregate is not a complete CBAM or TR-ETS monitoring, attribution, assurance or allocation methodology.
10. Online reproduction trusts the independently selected deployment anchor, frontend/source build and RPC; it is not a light-client finality proof.
11. Offline reproduction checks arithmetic and signatures only; it does not authenticate chain history.
12. A local or testnet deployment does not establish institutional participation, physical-phone measurement or real emissions reduction.

Source register

S01 / EU defaults annex / accessed 2026-10-21

S02 / TURKCIMENTO statement distributed by EQS / accessed 2026-10-21

S03 / CemNet industry reporting / accessed 2026-10-21

S04 / European Commission certificate prices / accessed 2026-10-21

S05 / CBAM amendment / accessed 2026-10-21

S06 / Original CBAM regulation / accessed 2026-10-21

S07 / DoCC board announcement / accessed 2026-10-21

S08 / ICAP pilot-design summary / accessed 2026-10-21

S09 / Official hackathon overview / accessed 2026-10-21

S10 / Official hackathon rules / accessed 2026-10-21

S11 / COP31 AI-Enabled Cleantech concept note / accessed 2026-10-21

S12 / circomlibjs Baby Jubjub source / accessed 2026-10-21

S13 / circomlib fixed-base scalar circuit / accessed 2026-10-21

S14 / Noble documentation / accessed 2026-10-21

S15 / CBAM calculation and attribution methods / accessed 2026-10-22

S16 / TR-ETS pilot procedures / accessed 2026-10-22

S17 / Noble ciphers documentation / accessed 2026-10-21

S18 / Noble hashes documentation / accessed 2026-10-21

S19 / Noble post-quantum documentation / accessed 2026-10-21

S20 / snarkjs setup and public powers-of-tau table / accessed 2026-10-21

S21 / Foundry installation and binary verification / accessed 2026-10-21

S22 / OpenZeppelin cryptography utilities / accessed 2026-10-21

S23 / IPCC 2006 Vol 2 Ch 2 stationary combustion / accessed 2026-10-22

S24 / IPCC 2006 Vol 3 Ch 2 mineral industry / accessed 2026-10-22

S25 / EU cement BAT conclusions, Decision 2013/163/EU / accessed 2026-10-22

S26 / IPCC cement production background paper / accessed 2026-10-22

S27 / NIST magnesium oxide / accessed 2026-10-22

S28 / NIST carbon dioxide / accessed 2026-10-22

S29 / NIST calcium oxide / accessed 2026-10-22

S30 / Ollama structured outputs / accessed 2026-10-22

S31 / Qwen2.5 1.5B model library / accessed 2026-10-22

S32 / ISO 14064-1:2018 public scope / accessed 2026-10-22

S33 / ISO 14064-3:2019 public scope / accessed 2026-10-22

S34 / ISO 14065:2020 public scope / accessed 2026-10-22

S35 / Wüst and Gervais, Do you need a Blockchain? / accessed 2026-10-22

S36 / Gazetted Türkiye Emissions Trading System Regulation / accessed 2026-10-22

S37 / DoCC announcement of pilot procedures / accessed 2026-10-22

Full source scopes and classes are retained in the JSON. References establish context; they do not endorse or certify this prototype.